

Grade 8
Unit 1
Lesson 5 Practice

Name Answer Key
Date _____
Period _____

Evaluate each expression using the order of operations.

$$\begin{aligned}
 1. \quad & (-2)^3 - 5|(-4.6 + 3.4)^2| - (-6.1) && (-2)^3 - 5|(-1.2)^2| - (-6.1) \\
 & && -8 - 5|1.44| - (-6.1) \\
 & && -8 - 7.2 + 6.1 \\
 & && -15.2 + 6.1 \\
 & && -9.1
 \end{aligned}$$

$$\begin{aligned}
 2. \quad & -5^2 + \left(\frac{1}{2}\right)^2 \div \frac{1}{20} \cdot (-7) && -25 + \frac{1}{4} \div \frac{1}{20} \cdot (-7) \\
 & && -25 + 5 \cdot (-7) \\
 & && -25 - 35 \\
 & && -60
 \end{aligned}$$

$$\begin{aligned}
 3. \quad & 14 - (-2^4 + 15) + 2| - 5.5| && 14 - (-16 + 15) + 2| - 5.5| \\
 & && 14 - (-1) + 2(5.5) \\
 & && 14 - (-1) + 11 \\
 & && 15 + 11 \\
 & && 26
 \end{aligned}$$

$$\begin{aligned}
 4. \quad & \frac{2}{5}(1 - .5)^2 - \frac{1}{2^3 + 6 \cdot 2} && \frac{2}{5}(0.5)^2 - \frac{1}{2^3 + 6 \cdot 2} \\
 & && \frac{2}{5}(0.25) - \frac{1}{8 + 12} \\
 & && 0.1 - \frac{1}{20} && 0.1 - 0.05 \\
 & && && = 0.05
 \end{aligned}$$

$$\begin{aligned}
 5. \quad & \frac{-2-10}{6^2-30} \div |1-3|^2 \\
 & \frac{-12}{36-30} \div |-2|^2 && -2 \div 4 \\
 & \frac{-12}{6} \div 4 && = -\frac{1}{2} \text{ or } -0.5
 \end{aligned}$$

$$\begin{aligned}
 6. \quad & 8 - |-4(3.75 + 6\frac{1}{4})| - 2^3 && 8 - 40 - 8 \\
 & 8 - |-4(10)| - 2^3 && -32 - 8 \\
 & 8 - |-40| - 2^3 && = -40
 \end{aligned}$$

$$\begin{aligned}
 7. \quad & -3.4(5^2 - 20) + (-2)^4 + |-11.2| \\
 & -3.4(25 - 20) + 16 + 11.2 \\
 & -3.4(5) + 16 + 11.2 \\
 & -17 + 16 + 11.2 \\
 & -1 + 11.2 && = 10.2
 \end{aligned}$$

$$\begin{aligned}
 8. \quad & \frac{(-3)^2 - 3^2}{-2.2 \cdot 4|-3-5|} \\
 & \frac{9-9}{-2.2 \cdot 4|-8|} \\
 & \frac{0}{-2.2 \cdot 32} \\
 & 0
 \end{aligned}$$

$$\begin{aligned}
 9. \quad & 6 - 3\frac{1}{2}|-4^2 - (-7)| + 2.6|(-3)^2| \\
 & 6 - 3\frac{1}{2}|-16 + 7| + 2.6|9| \\
 & 6 - 3\frac{1}{2}|-9| + 2.6(9) \\
 & 6 - 31.5 + 23.4 \\
 & -25.5 + 23.4 \\
 & -2.1
 \end{aligned}$$

10. Create and evaluate a numeric expression with up to six operations.

Answers will vary.